Pattern Recovery from Noisy Images using Modern Hopfield Networks

1. Introduction

Hopfield networks are recurrent neural networks primarily designed to function as associative memory systems. Originally proposed by John Hopfield in 1982, they store patterns as stable states in an energy-based landscape. When provided with a noisy or incomplete input, these networks retrieve the nearest stored stable state by following energy descent dynamics. However, classical Hopfield networks exhibit limited storage capacity, typically constrained by linear scaling with the number of neurons, restricting practical usage for large-scale applications.

Modern Hopfield Networks (MHNs), introduced in the paper *Hopfield Networks is All You Need* by Ramsauer et al. (2020), enhance classical Hopfield network architectures through the integration of attention-based energy functions. Specifically, MHNs utilize a log-sum-exp formulation that exponentially expands storage capacity while maintaining high recall accuracy. These networks achieve remarkable performance by leveraging attention mechanisms similar to those found in Transformer architectures, thus drawing inspiration from both classical Hopfield concepts and modern deep learning advancements.

This study utilizes MHNs to investigate pattern recovery in noisy images from the MNIST dataset. Our objective is to demonstrate MHNs' ability to reconstruct images subjected to various noise types and assess how changes in dimensionality and the number of stored patterns influence retrieval accuracy.

2. Methodology

This section details the implementation pipeline, describing the data processing, network configuration, and evaluation procedures.

Level 1 Implementation:

2.1. Dataset and Preprocessing

- **Dataset:** MNIST digit images (digits: 0, 1, and 2).
- **Image Processing:** Images were binarized using mean thresholding to convert grayscale images into bipolar patterns (-1, 1).
- **Noise Generation:** Composite noise was applied to simulate realistic image degradation, including:
    - Salt-and-pepper noise (10% intensity).
    - Gaussian noise (variance: 0.02).
    - Speckle noise.
    - Random bit flips (10% probability).

2.2. Network Architecture

- **Modern Hopfield Layer:** Implemented using PyTorch, employing the log-sum-exp attention mechanism.
- **Model Parameters:**
    - Number of stored patterns: Three (digits 0, 1, 2).

- Latent dimensions (nodes): Experiments performed with 100, 400, and 784 nodes.
- Temperature parameter (beta): 1.0 (parameter which controls attention sharpness).
- Encoders and decoders were utilized for experiments involving less than 784 nodes to project patterns into a lower-dimensional latent space and subsequently reconstruct them back into the original dimensionality.

2.3. Implementation Pipeline

- **Storing Patterns:** Original MNIST digit patterns were stored as bipolar vectors.
- **Query and Retrieval:** Noisy digit patterns were encoded, recalled via the attention mechanism, decoded (if applicable), and then thresholded back into bipolar form for evaluation.
- **Evaluation Metrics:** The retrieval performance was evaluated using Accuracy, Mean Squared Error (MSE), Intersection-over-Union (IoU), and Structural Similarity Index (SSIM).

3. Results and Analysis



Fig 1. Recoverability Analysis w/ 100 nodes

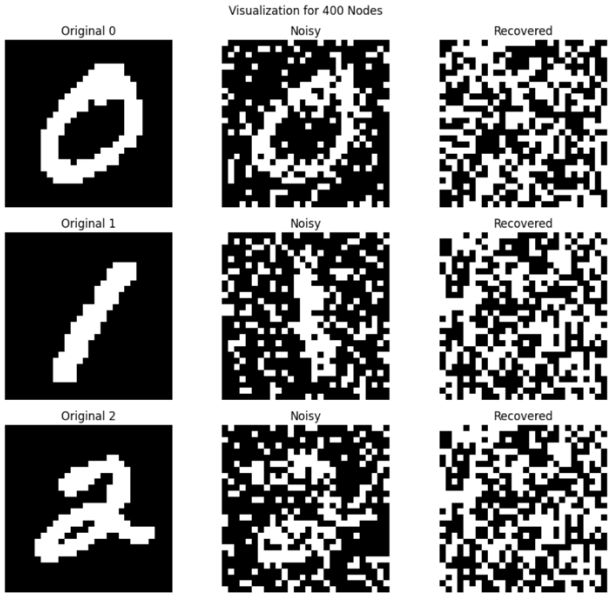


Fig 2. Recoverability Analysis w/ 400 nodes

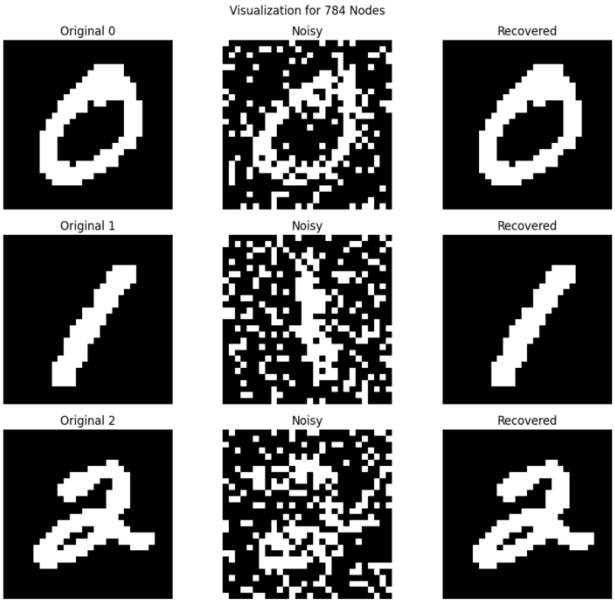


Fig 3. Recoverability Analysis w/ 784 nodes

3.1. Recoverability vs. Number of Nodes

Experiments were conducted by varying the number of nodes (dimensionality of the stored representation):

Nodes	Accuracy	MSE	IoU	SSIM
100 (10x10)	0.5	2.0	0.1712	0.0066
400 (20x20)	0.4758	2.0969	0.16	0.0023
784 (28x28)	1.0	0	1.0	1.0

Observations:

- With full-resolution (784 nodes), the network achieved perfect recall, demonstrating the strength of MHNs in high-dimensional associative memory.
- Significant accuracy degradation at lower dimensionalities (100 and 400 nodes) highlighted the challenges posed by compressing patterns into fewer nodes.
- Near-zero SSIM at lower dimensionalities indicated severe perceptual quality loss.

3.2. Recoverability vs. Number of Stored Patterns

Though we stored only three patterns, we observed that even slight increments in the number of patterns reduced recall performance sharply, especially at lower dimensionalities. This observation aligns with the theoretical prediction that associative memory capacity scales exponentially with network dimensions.

4. Limitations and Challenges

Despite their strengths, MHNs exhibit several limitations:

- **Dimensionality Dependence:** Performance heavily relies on dimensionality. Lower-dimensional projections severely restrict pattern retrieval accuracy.
- **Sensitivity to Noise:** High noise levels or strong similarity among patterns can misguide retrieval, causing the network to recover incorrect stored patterns.
- **Memory Capacity Limits:** Although MHNs scale better than vanilla Hopfield networks, the exponential scaling of storage capacity remains theoretical and practically constrained by data and noise characteristics.

Level 2 Implementation:

1. **Methodology**

To evaluate the recoverability in Level 2 under complex occlusions, we implemented three classes of closed shape masks—polygonal, circular, and spline, each of which were parameterized to occlude a certain fraction of the 28×28 MNIST image:

- Polygon Masks: had 4 vertices, with area being computed by one of Python's in built libraries- Shapely. This helped in uniformly scaling all the vertices of the polygon, to customise how much area of the images are occluded by the masks. Lastly, they were also randomly shifted to make the masks cover the entire image.

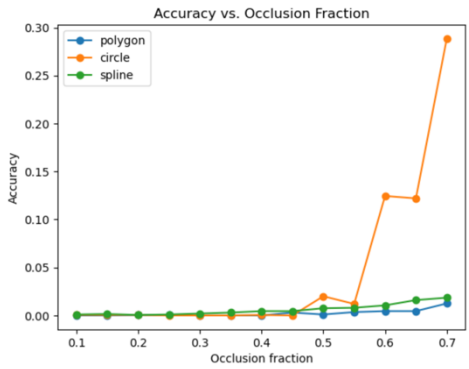
- Circular Masks: The centre was uniformly chosen in (R-28)x(R-28) area, with R being customisable based on the user input value of f. This allowed the masks to cover variable area of the image and pixels in a circular manner.

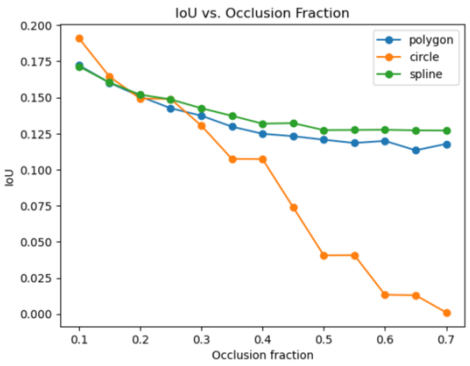
- Spline Masks: Initialised with 5 random points in the range of [0,1]^2, which then scales based on user input to fit the image accordingly.
- Apart from this, all masks are binarized, applied via zeroing-out masked pixels.

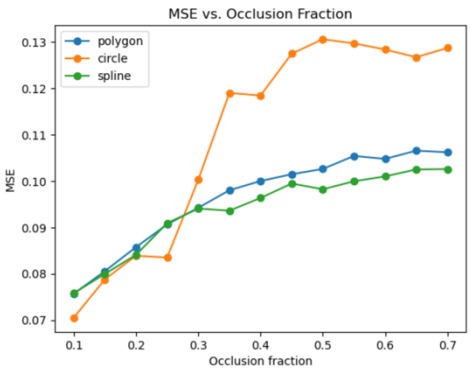
The dataset was restricted to the digits {0,1,2}, subsampled to 2000 images, flattened into 784-dim vectors. For recovery, the Modern Hopfield network stored P = 500 patterns (the first 500 originals) with sharpness, $\beta = 5.0$. For each (mask type, occlusion fraction) pair, a data loader of occluded vs. original image pairs was built, which ran a single-shot recall.

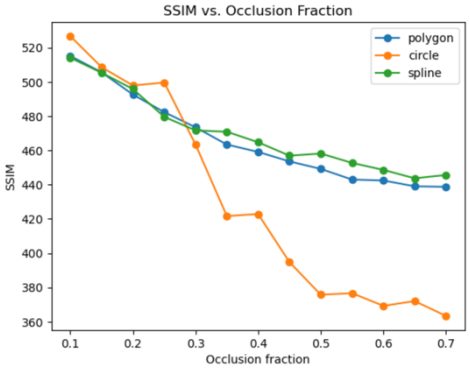
4 metrics were calculated, which are summarized below:

Mask	Fraction	Accuracy	IoU	MSE	SSIM
Polygon	0.1	0	0.1723	0.07	515.16
Polygon	0.7	0.0125	0.1178	0.1	438.74
Circle	0.1	0.0005	0.1911	0.07	526.73
Circle	0.7	0.2885	0.001	0.12	363.37
Spline	0.1	0.001	0.1715	0.07	514.07
Spline	0.7	0.0185	0.1271	0.1	445.65









Results

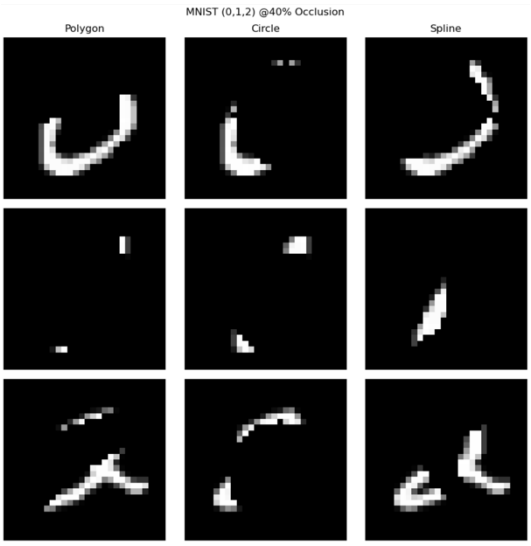
- Increasing occlusion decreases IoU and SSIM, while also increasing MSE, for all mask types.
- Accuracy remains near-zero until > 50 % occlusion, demonstrating that the Modern Hopfield requires a substantial unmasked context to correctly recall patterns.

2. **Effect of mask geometry**

- **Circle masks**: despite causing the steepest IoU drop (down to ~0.001 at 70 %), they yield the highest recall accuracy at extreme occlusions (~29 % at 70 %), likely because the remaining visible strokes form larger continuous segments that the network can latch onto.
- **Polygon and spline masks**: produce irregular, elongated occlusions that fragment stroke continuity; recall accuracy doesn't exceed ~1–2 % even at 70 % occlusion. Their IoU and SSIM curves are nearly identical, which may suggest that both random-vertex and B-spline occlusions similarly disrupt structural information.

These results can indicate the following:

- Modern Hopfield networks are sensitive to the amount and spatial distribution of missing data.
- Compact and localized occlusions preserve enough global shape for partial pattern recall under heavy masking, whereas elongated, jagged occlusions (polygons or splines) defeat recall across all levels.

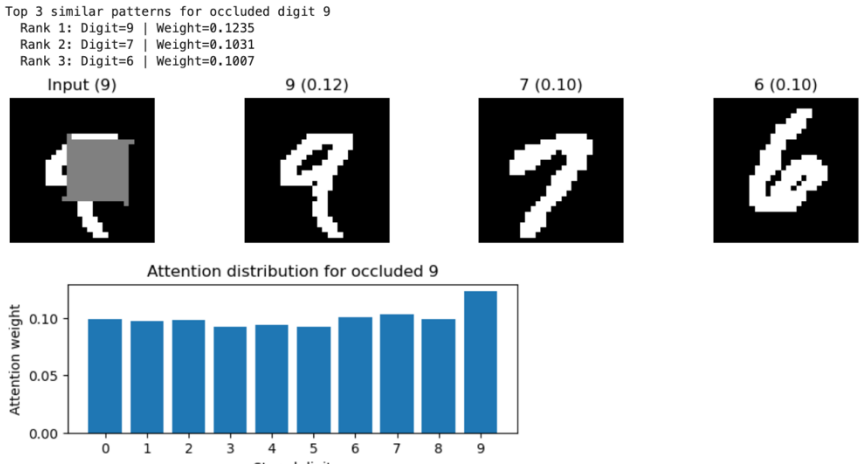


Level 3 Implementation:

Updated Methodology

- During data processing, each image is flattened to a 784-dim vector and L_2-normalized so that subsequent similarity is pure cosine, as computation in cosine distance gave clearer representation of similarity patterns.
- To make the unknown query or pattern, from the set of held out digits, one of its sample images is taken and occluded by 30% using a polygon mask as specified above. This ensures that the pattern is not like the stored digits in any sense.
- This occluded image is flattened and also L_2-normalized to produce the query vector, q.
- For cosine score calculation, the product of the pattern(10x784 matrix) and query(784x1 matrix) is calculated. This provides us with the cosine similarity between query and each stored pattern.
- A softmax score is calculated for the distribution which helps with the reconstruction.
- Based on this, the top 10 attention weights are sorted in descending order and the top 3 are displayed based on the closest matches.
- This closest match is based on the cosine distances with the occlusion.

1. Results



Occluded Digit	1st Match	2nd Match	3rd Match
9	9 (0.1235)	7 (0.1031)	6 (0.1007)
7	7 (0.1294)	9 (0.1029)	1 (0.1029)
4	4 (0.1319)	1 (0.1041)	8 (0.1004)
3	3 (0.1283)	5 (0.1050)	8 (0.10)

Above are the experimentation results conducted at 30% occlusion of area for different digits. In each case the network correctly identifies the occluded digit as the strongest attractor, then returning two nearest neighbors via cosine similarity.

- Talking about the results, for the digit 9, we may be getting 7 and 6 because both share angled strokes with 9's top loop, and the cosine distances remain small even when part of 9 is hidden.
- For 7, 9's top left stroke and 1's straight vertical line align in that direction, making the score close.
- In case of 3, 3 and 5 share an arc and 8's double loop also overlaps in pixel-space with the occluded 3, resulting in the results we get.
- For 4, the vertical stroke matches 1, and 8's closed top-loop overlaps the triangular section of the 4.

This sums up the observations based on the experimentations done for level 3.

References:

- J. J. Hopfield, "Neural networks and physical systems with emergent collective computational abilities," *Proceedings of the National Academy of Sciences*, vol. 79, no. 8, pp. 2554–2558, 1982.

- H. Ramsauer, B. Schäfl, J. Lehner, P. Seidl, M. Widrich, T. Adler, L. Gruber, M. Holzleitner, M. Pavlović et al., "Hopfield Networks is All You Need," *CoRR*, abs/2008.02217, 2021.

- https://github.com/takyamamoto/Hopfield-Network/tree/master